

Will Maberry

 will-maberry |  willmaberry.com |  will-maberry

RESEARCH INTERESTS

Computational epidemiology; spatiotemporal modeling; infectious disease forecasting; rare-event detection; One Health; biosurveillance; geospatial ML

EDUCATION

Mississippi State University

M.S. Vet. & Biomed. Science Aug. 2026 – May 2028

Computational Biology Concentration

Research Advisor: Dr. Isaac Jumper

The University of Texas at Arlington (UTA)

B.S. Computer Science Aug. 2022 – May 2026

HONORS & CERTIFICATIONS

- Summa Cum Laude: UTA (GPA 3.824)
- Judge's Choice Award: UTA College of Engineering Innovation Day (2026), for 2025 internship research
- Fundamentals of AI Certificate: UTA

TECHNICAL SKILLS

Forecasting & Modeling

Spatiotemporal forecasting; automated leakage-safe ML pipelines; rare-event modeling; ablation studies

Machine Learning

PyTorch; TensorFlow; Scikit-learn; Imbalanced-learn; Optuna; AutoGluon

Geospatial & Scientific Computing

GeoPandas; Google Earth Engine; raster processing

Programming & Data Engineering

Python; C; Java; SQL; Bash; Pandas; NumPy

Pipelines & Deployment

GitHub/Git, FastAPI; Flask; Docker

Visualization

Matplotlib; Folium/Leaflet; interactive dashboards

RELEVANT COURSEWORK

Machine Learning & AI

Machine Learning (ML); Neural Networks & Deep Learning; Artificial Intelligence (AI); Computer Vision

Data & Algorithms

Data Analysis & Modeling Techniques; Design & Analysis of Algorithms; Algorithms & Data Structures

Mathematics

Calculus I–III; Linear Algebra; Discrete Structures; Theoretical Computer Science

Graduate Coursework (In Progress)

Applied Veterinary Epidemiology; Management of Commercial Layers; Statistical Methods

EXPERIENCE

Mississippi State University

Graduate Research Assistant

Aug. 2026 – Present

USDA ARS NACA Fellow

- Developing a case-control study of HPAI biosurveillance practices and outbreak-associated factors.

- Planned analyses will use USDA ARS's ATLAS HPC system to process and analyze study data.

USDA Agricultural Research Service

Undergraduate Research Assistant

Jul. 2025 – Jul. 2026

- Built a **retrospective, operational simulation** county-month HPAI forecasting framework for the U.S., enforcing temporal constraints to replicate deployment conditions.

- Evaluated **40 spatiotemporal and gradient-boosting model configurations** leveraging county adjacency networks across a 24-month backtest, spanning **74,616 county-month forecasts**.

- Achieved **73% balanced accuracy and 74% recall** against 0.8% outbreak-positive prevalence, successfully predicting 417 of 563 outbreak-positive county-months.

USDA Agricultural Research Service

Research Intern

May 2025 – Jul. 2025

- Developed national HPAI classification and pseudo-forecasting models using future climate covariates; achieved **75%+ balanced accuracy** on 2022–2024 data (~1% positive).

- Presented findings to the **USDA ARS Chief Scientist** and National Program Leaders.

OpenAI

Engagement Manager

Jun. 2024 – Jun. 2026

- Scaled OpenAI's official Discord community to **850k+ users** (680% growth) through technical programming, online events, and community engagement initiatives.

TEACHING & COMMUNITY

- **Teaching Assistant (Operating Systems):** Supported 120+ students through office hours and review sessions.

- **OpenAI Community Volunteer:** Supported global AI education initiatives and technical discussions (2022–2024), later transitioning to a paid contract role.

- **Association for Computing Machinery (ACM):** Education Director for 1500+ member organization; led 15+ professional and educational workshops.

- **The Wesley Foundation @ UTA:** Sole student representative on Board of Directors overseeing \$100k+/year budget.

PUBLICATIONS

Blackstock, J., Maberry, W., et al.

“A forward-looking high pathogenicity avian influenza monthly risk model for counties in the conterminous United States”

Manuscript submitted to Scientific Reports (May 2026)